



No Coding Required!

Build AI Automation Workflows with Visual Tool

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Background

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The Challenge:

Using AI to automate in non-technical environments

01

THE OPPORTUNITY

Using AI for automation can greatly improve productivity at work

02

THE PROBLEM

But in non technical environments (administrative teams), very few people even want to touch code

03

THE GOAL

So we use AI to give people who can't code — or have no technical background the right tools to build their own AI automation workflows, and save themselves time

My solution:

n8n — an open source GUI Automation Tool



<https://github.com/n8n-io/n8n>

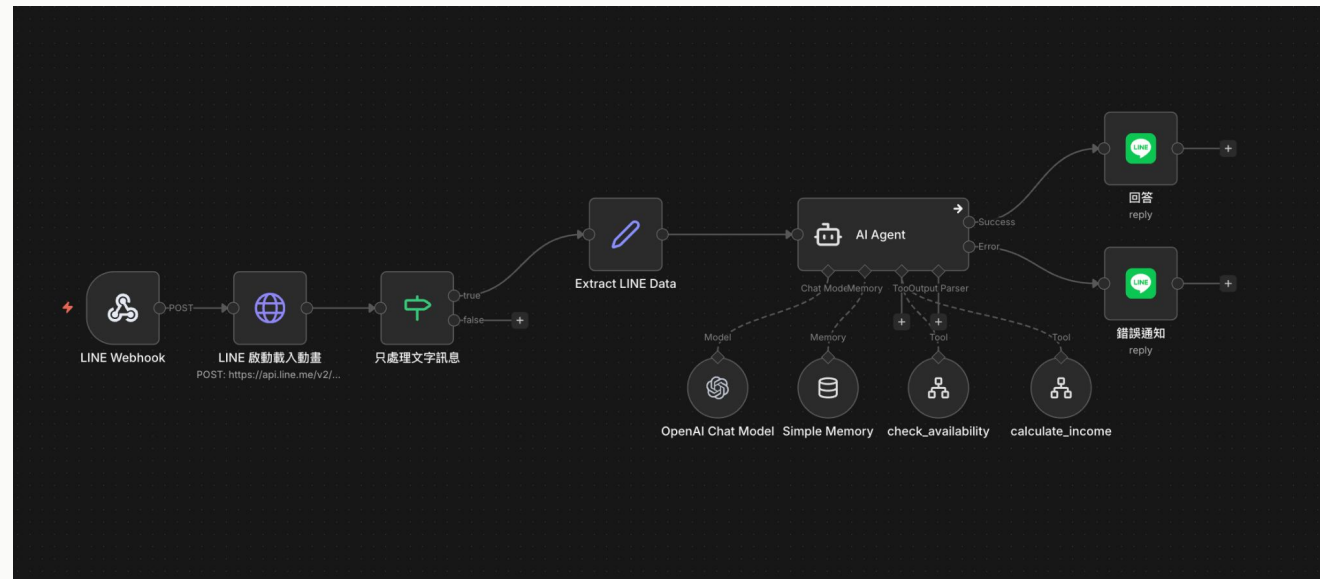
What is n8n?

A “workflow automation” tool

Connects different services together, like Lego bricks

No coding needed — just drag and connect

Google Drive, LINE, YouTube — all can connect



n8n Deployment Options

Two deployment methods and their differences

Local Deployment (Docker)

Data storage location

Your own computer or laptop

Can it run 24H?

Only works while the computer is on;
stops if shut down

Webhook (external) receiving

Firewalls will block it; extra setup needed if required

Cost

Completely free

Setup difficulty

Moderate — needs some technical skill

Cloud Deployment (cloud platform)

Data storage location

Cloud server data centers online (AWS or GCP)

Can it run 24H?

Runs continuously — no need to keep your own computer on

Webhook (external) receiving

Platform provides a secure URL (HTTPS) — receives real-time webhooks perfectly

Cost

Usage-based, roughly US\$10–20 per month

Setup difficulty

Beginner-friendly — as easy as online shopping

Visual nodes: the benefits for people who can't code

Automate real work without writing code — and keep it visible and maintainable



See every step

See where data flows and where it breaks
node status wrong make it obvious, no
log-digging required



The flow is the documentation

Colleagues grasp the logic from the diagram
and can take over
no more “code only you understand”



Fill in fields, no syntax

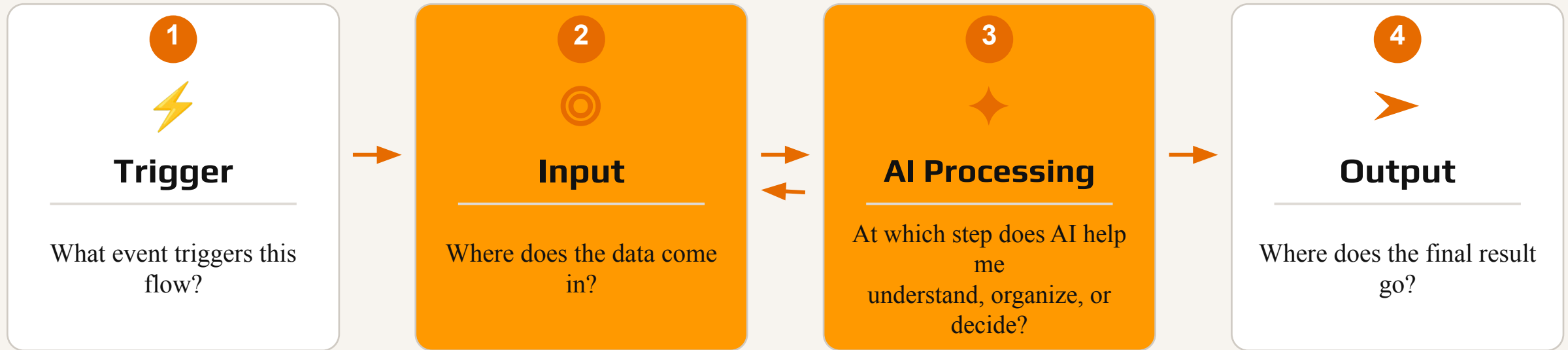
Configuring a node means filling in fields
no memorizing syntax, no fear of typos



Change one part without breaking everything

Want a tweak? Just drag a node
no worry about affecting other parts

T-I-A-O Flow Breakdown



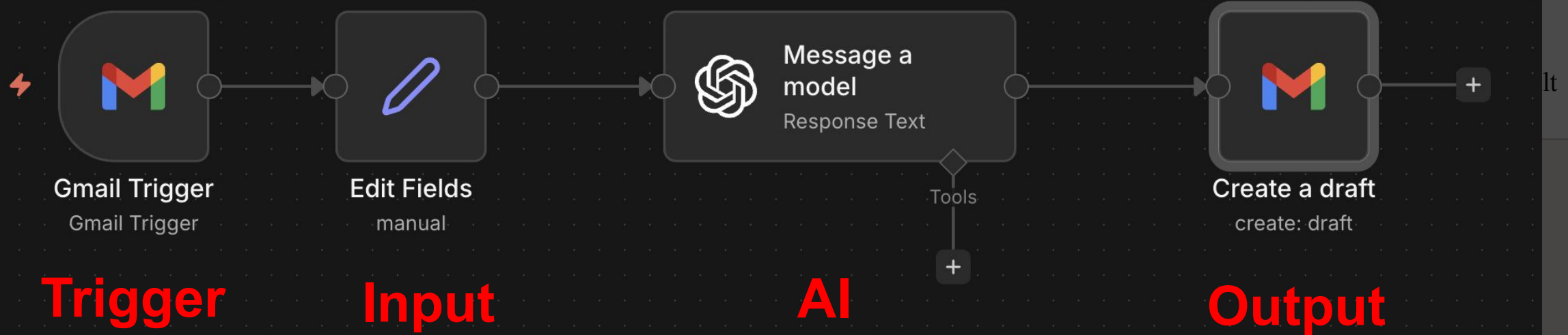
This is the biggest benefit of visual workflows for non-engineers:

It turns abstract programming logic into a flow you can see, test, and maintain



T-I-A-O Flow Breakdown

After an email arrives, AI auto-organizes it into a summary and draft



Before building, have AI organize your ideas and produce the architecture

01

Describe the problem first

Explain how you do it now, how tedious it is, and what you want it to become

prompt:

I want to automate a repetitive task
Don't rush to a solution — clarify the problem first

How I do it now: _____

What triggers it: _____

How long it takes each time: _____

What I want it to become: _____

Tools involved: _____

First restate my needs in your own words
then ask me the 3 most critical questions

02

Ask AI to design the architecture

AI gives you a draft flow
you just judge if it's right

prompt:

Based on the needs above, design an n8n workflow for me

One node per line: node name? what does this step do?

Break down the steps using the "T-I-A-O framework":

[T] Trigger: what event starts this flow?

[I] Input: Where does data come from?

[A] AI Processing: which step needs AI?

[O] Output: where should the final result go?

I'm not an engineer — please explain in plain language

03

Build it piece by piece

Test after each step
Stuck? Paste the error to AI

04

Ask AI to optimize

ask AI where it might fail
or what could be optimized

prompt:

This flow is already running
help me find what conditions would break it

for each risk, give a fix that can be handled within n8n

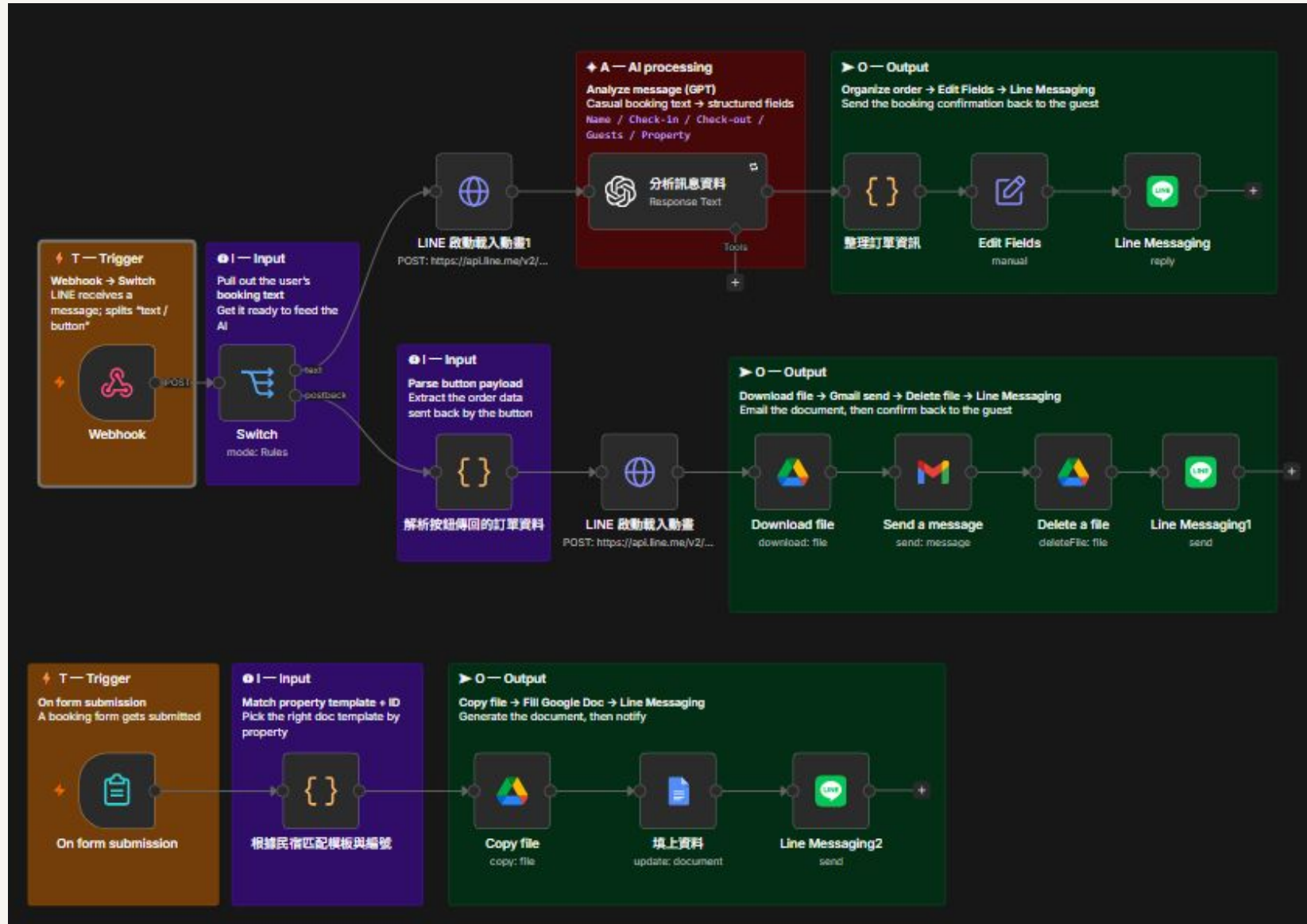
Use three columns: "Risk → Impact → Fix", starting with the most severe



DEMOS



Booking Form Automation



Standard LLM node prompt: User | System

An AI node has two fields — understand what goes in each



User Prompt

The data to process / the immediate instruction to AI

The data the user inputs each time
text, instructions, data variables

```
{{ message from LINE }}
```

```
Please analyze for me..
```



System Prompt

Role / Task / Rules / Examples / Output format

Tell AI “who you are, what rules to follow,
give examples, how to output”

Role You are a professional personal-finance and bookkeeping assistant.

Task Take the user’s casual spending notes and automatically convert them into a standard bookkeeping format.

Rules

Force each expense into 5 categories: Food, Transport, Shopping, Entertainment, Other.

Precisely extract the “item” and the “amount (digits only)”.

If the user doesn’t mention a time, auto-fill the date as “today”.

Example User input: “Just bought coffee and a sandwich at 7-11 for 120” Output: Date: today Category: Food Item: coffee and a sandwich Amount: 120

Output format Plain text. No blank lines or any Markdown syntax, and no greetings.

Standard LLM

An AI node has



User Prompt

The data to process
to AI

The data the user inputs e
text, instructions, data var

```
{{ message from LINE
```



Standard LLM | Booking bot

Role / Rules / Examples / Output format

Role

You are a professional booking assistant; extract booking info from messages.

Task / Fields

Extract: Name / Check-in / Check-out / Guests / which property

Rules

- Normalize dates to YYYY/MM/DD
- Leave missing info blank – don't make it up
- This year is 2026 (messages often omit the year)

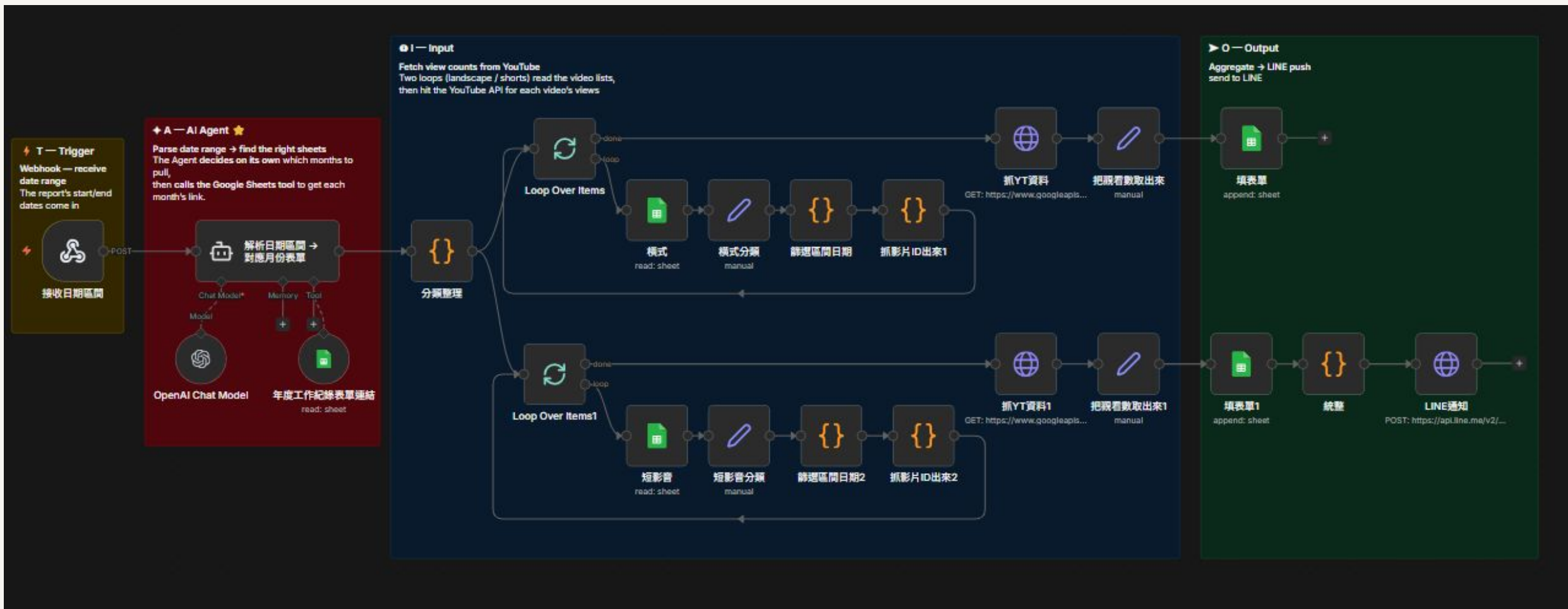
Example

"3/19-3/26, 2 guests, Ming Wang, 0900-000-000"

Output format

Return JSON only, no Markdown.

YouTube × Google Sheet: Automatic View-Count Aggregation



AI nodes: Standard LLM | AI Agent

Both connect to AI, but “whether it uses tools on its own” decides which to use



Standard LLM

Text input



LLM parses

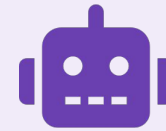


Output

- AI does one thing only: analyze or process your data
- Doesn't touch external data or take extra actions
- Fixed input → stable, predictable output



My case | Booking bot: one spoken booking →
extracted into structured fields



AI Agent

Text input



Agent reasons



Output



Calls a tool to query Google Sheets

- AI decides on its own: whether to look up data / use tools
- Can call tools and do multi-step reasoning
- Good for tasks that need “using external tools”



My case | View-Count Aggregation understand the
date range → look up the table for the matching link

AI nodes: Standard LLM | AI Agent

Both connect to AI,

which to use



Stan

Text input

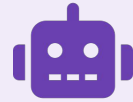


- AI does one thing on data
- doesn't touch external
- Fixed input → stable



My case | Guest

→ extracted into s



AI Agent | View-Count Aggregation

Key point: teach it to “use tools”

Role

You are the view-count aggregation assistant ; find the matching month links by date range.

Available tools

“Annual work-log sheet link” (Google Sheets)

When to use the tool

Whenever the user gives a date range, you must call the tool to query the table – never guess from memory.

Rules

1. Use the year (2026) to determine which months are covered
2. Call the tool to get each month's link
3. Return all months – don't miss a single one

Safeguards

- No link found → fill that month with empty string “”, don't fabricate
- Tool fails twice in a row → report directly, don't loop forever

Output format

Return JSON only, no Markdown.

ent

agent
asons



Output

Is a tool to query Google Sheets

Whether to look up data /

Multi-step reasoning

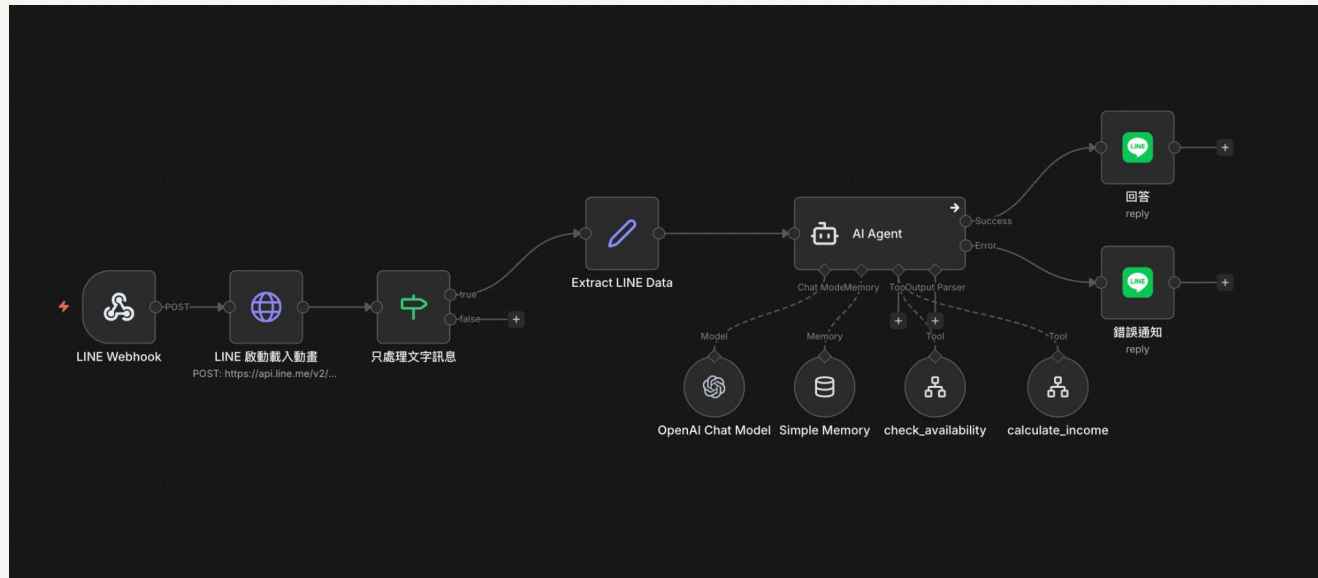
“using external tools”

report: understand the date
ole for the matching link

International Video Subtitle Factory



Questions?



<https://github.com/n8n-io/n8n>

Thank you